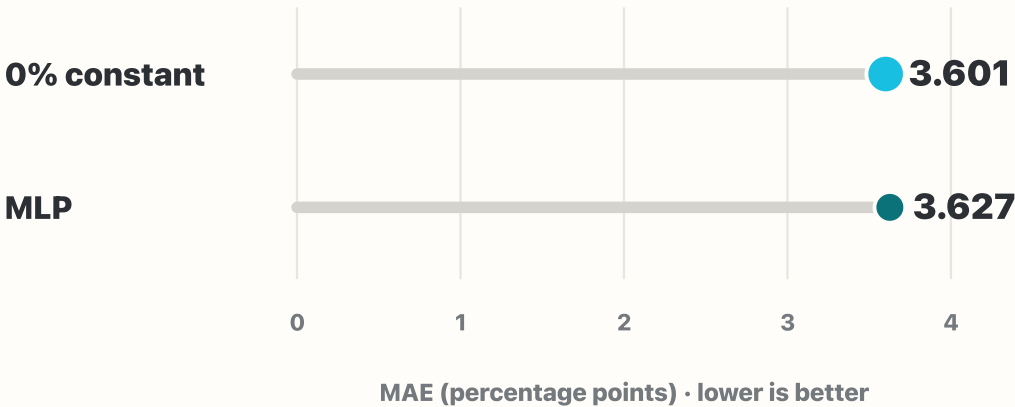
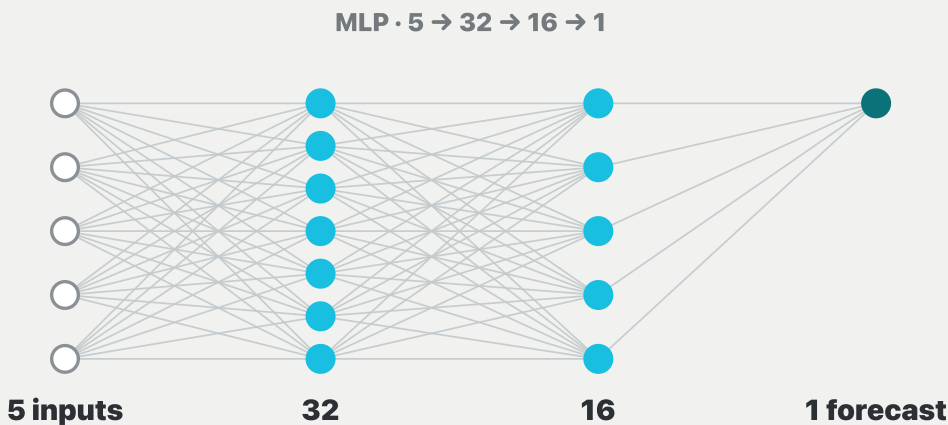


I SPENT 24 HOURS ON A NEURAL NETWORK. ALL I GOT WAS THIS LOUSY T-SHIRT.

My fancy model can learn very complex functions.
"Predict 0% growth" still beat it on average absolute error (MAE).

TARGET: NEXT-QUARTER DOMESTIC DEPOSIT GROWTH

Deposits	\$100m → \$96m
Actual growth	-4%
Zero forecast	0%
Absolute error	4 percentage points



THE CONSTANT PREDICTED 0% GROWTH FOR EVERY BANK.

DEPOSITS CAN LEAVE BEFORE LOANS PAY BACK.

EXAMPLE: \$100 IN ASSETS. \$10 CASH. \$90 LOANS.



CASH IN THE VAULT

\$10



MONEY ALREADY OUT AS LOANS

\$90

A customer withdraws \$15. Cash is \$5 short.

THE BANK'S TREASURY TEAM MUST FIND THE CASH.

HOW MUCH CASH IS ENOUGH?



There is no single cash percentage for every
U.S. bank.

Banks must manage withdrawals. Firms under the
full liquidity coverage rule hold liquid assets
covering **100% of 30-day stressed net cash
outflows.**

CASH

DEPOSITS

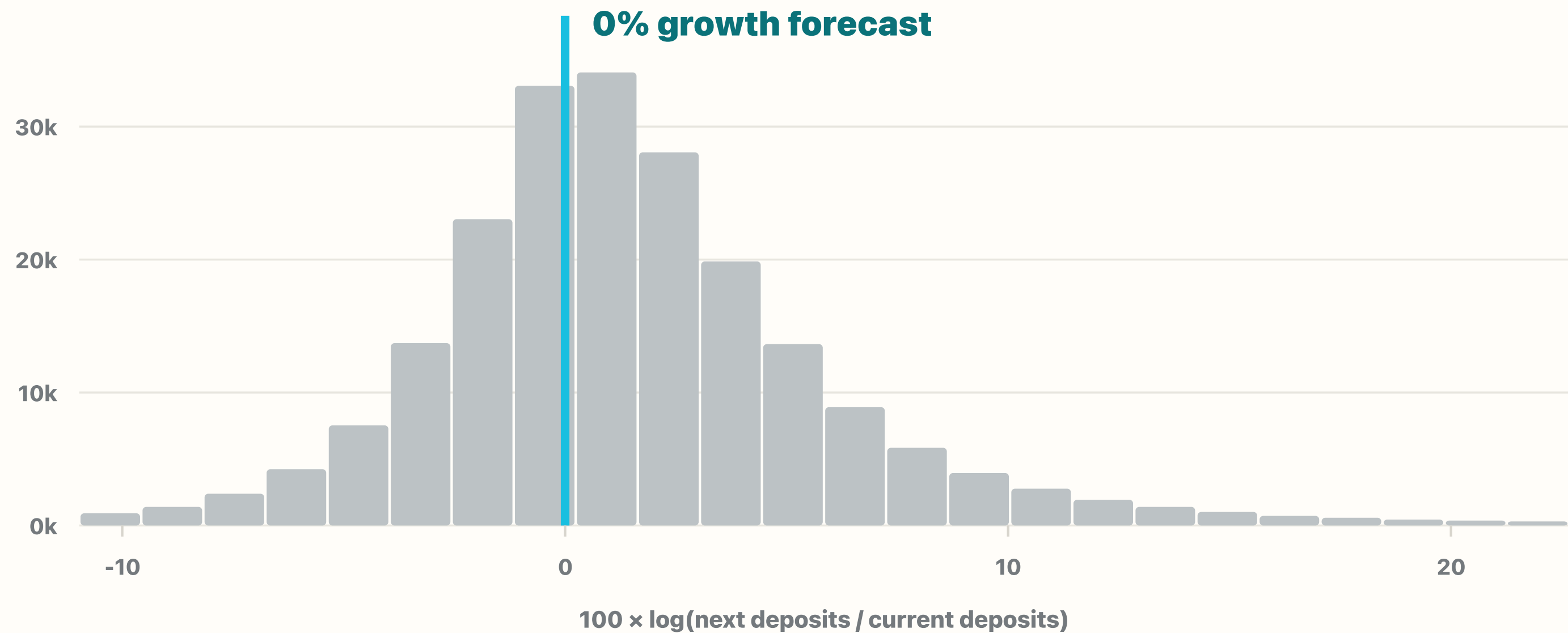
LOANS

CAPITAL

**CUSTOMERS DEPOSIT AND WITHDRAW MONEY.
I'M FORECASTING THE NET CHANGE NEXT QUARTER.**

Example only. Reserve requirement ratios are 0%; separate liquidity rules vary by bank. Liquid assets include eligible securities, not just cash. Source: Federal Reserve, Regulation WW. This project forecasts deposit growth, not bank failure.

SO WHY WAS 0% GROWTH SO HARD TO BEAT?



A LOT OF THE TRAINING TARGETS LIVE NEAR ZERO.

Central 98% shown · full tails stayed in the experiment · display scale is 100 × log growth

BIG ERRORS GET **SQUARED**. SO THEY CAN PULL THE MODELS AWAY FROM 0%.

An error 5× as large gets 25× the squared-error penalty.

FIXED BENCHMARK

0% BASELINE

$$\hat{y} = 0$$

Always predicts 0%.
It never adjusts to reduce an error.

ILLUSTRATIVE TRAINING ERRORS

2-unit error $\rightarrow 2^2$

10-unit error $\rightarrow 10^2$

4

VS

100

BIG ERRORS COST MUCH MORE.

LEARNED MODELS

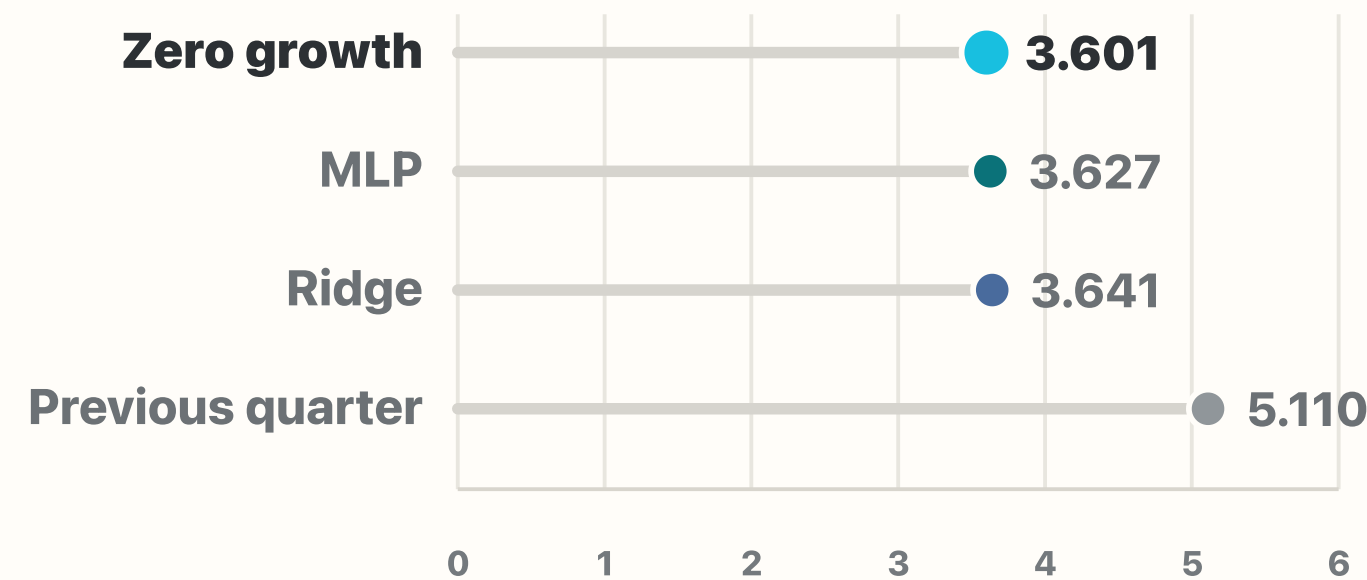
RIDGE + NEURAL NET

Can move away from 0% to reduce
those expensive large training errors.

**A LOWER SQUARED ERROR CAN COME WITH
A SLIGHTLY HIGHER AVERAGE ABSOLUTE ERROR.**

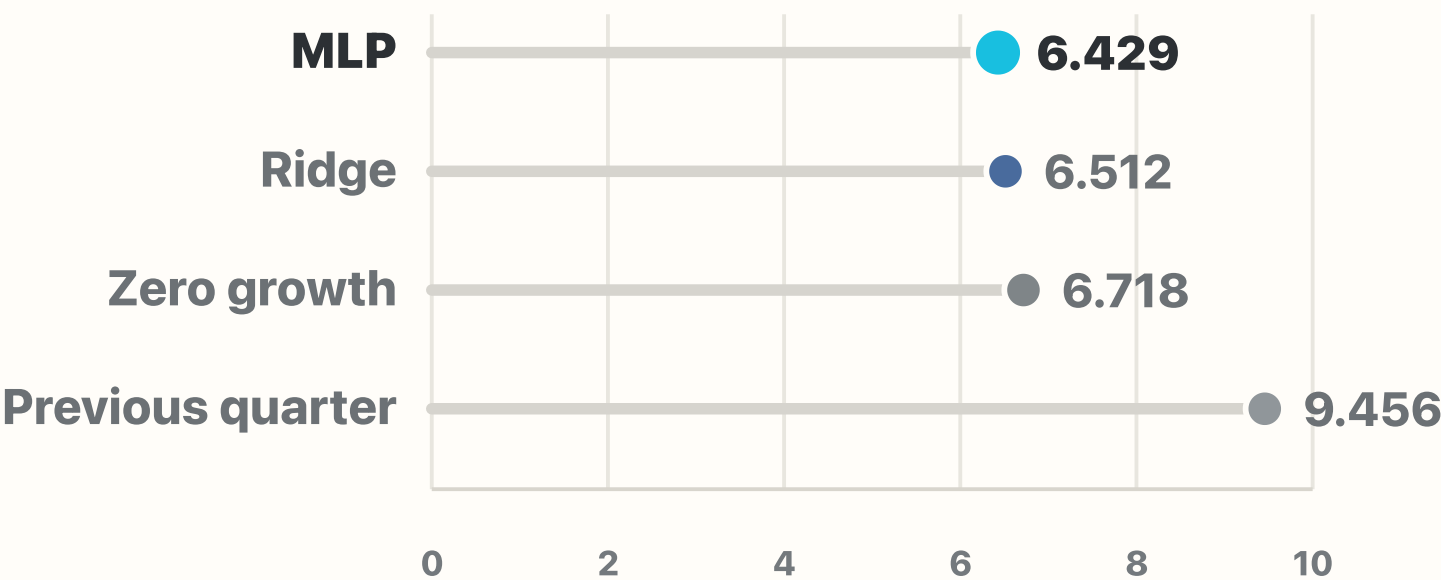
ZERO WON MAE. THE NEURAL NETWORK WON RMSE.

AVERAGE ABSOLUTE ERROR MAE



MAE (percentage points) · lower is better

LARGE ERRORS COUNT MORE RMSE



RMSE (percentage points) · lower is better

“WINNER” DEPENDS ON THE ERROR METRIC WE CHOOSE.
MAE averages absolute errors. RMSE squares errors first, so large errors count more.

0% TIES EVERY BANK.

THE LEARNED MODELS TELL ME WHO TO REVIEW FIRST.

I am not a financial advisor. I do have a sorted table.
In this March 2024 queue, review Young Americans Bank first: Ridge predicts the largest deposit decline, −4.02%.
Check its reports to understand the forecast; this is not a current distress warning.

ORDERED REVIEW QUEUE · SIMPLE LINEAR MODEL (RIDGE) · FIRST 10 OF 454

RANK	BANK	PREDICTED GROWTH	0% BASELINE
01	YOUNG AMERICANS BANK	-4.02%	0% · tied
02	PIONEER COMMERCIAL BANK	-3.32%	0% · tied
03	SUMITOMO MITSUI TR BK U SA	-2.83%	0% · tied
04	OLD GLORY BANK	-2.75%	0% · tied
05	TURTLE MOUNTAIN STATE BANK	-2.11%	0% · tied
06	COLUMBUS STATE BANK	-2.05%	0% · tied
07	FDS BANK	-2.03%	0% · tied
08	BOELUS STATE BANK	-1.96%	0% · tied
09	LEMONT NATIONAL BANK	-1.80%	0% · tied
10	CITIZENS NB OF CROSBYTON	-1.75%	0% · tied

GOAL: FIND MORE LOWEST-GROWTH BANKS WITH THE SAME 454 REVIEWS. HIGHER IS BETTER.

Random selection

45.4 expected

Linear model (Ridge)

112 found

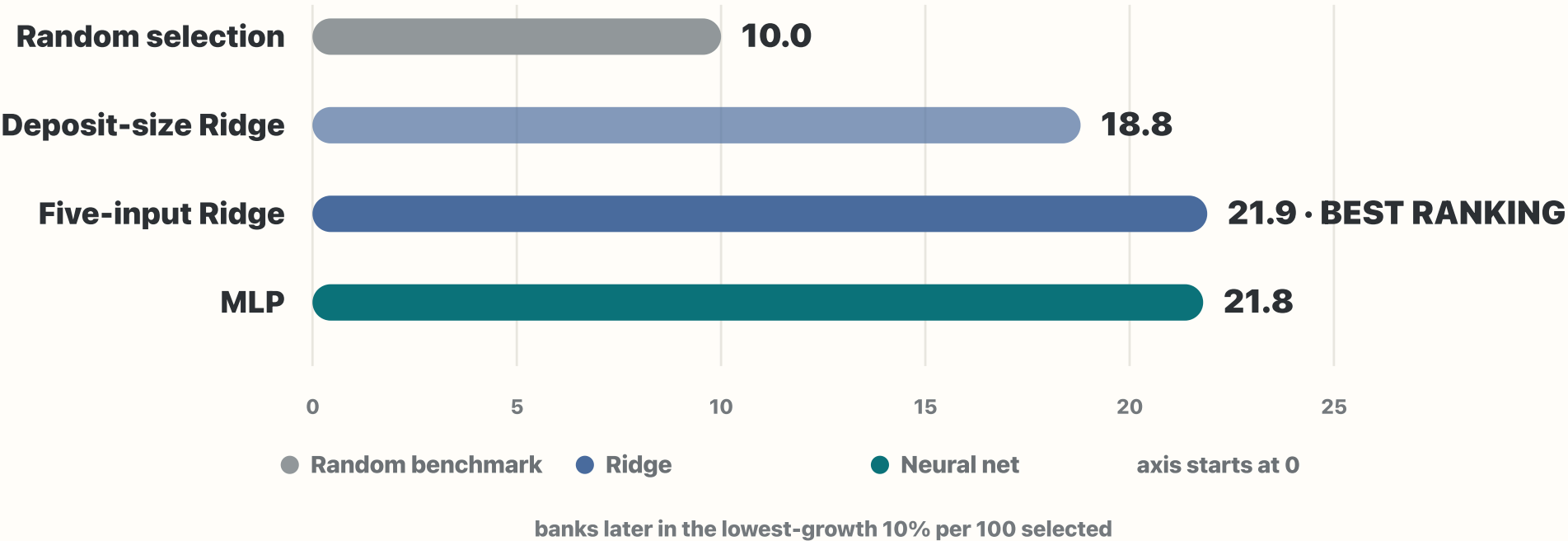
Neural network

109 found

4,536 eligible banks · March → June 2024. “Lowest-growth” = later in that quarter’s bottom 10% of deposit growth.
112 matches across the full linear-model queue; 109 for the neural net. Review priority is not a distress diagnosis.

I THREW OUT 4 INPUTS. KEPT ONLY DEPOSIT SIZE.

I tested deposit size alone to see whether four extra inputs actually improved the review list.



MAE WINNER: PREDICT 0% → 3.601. Deposit-size Ridge: 3.654 · Five-input Ridge: 3.641 · Neural net: 3.627.
All three lost to zero on MAE. Zero ties every bank, so it cannot order a review list.

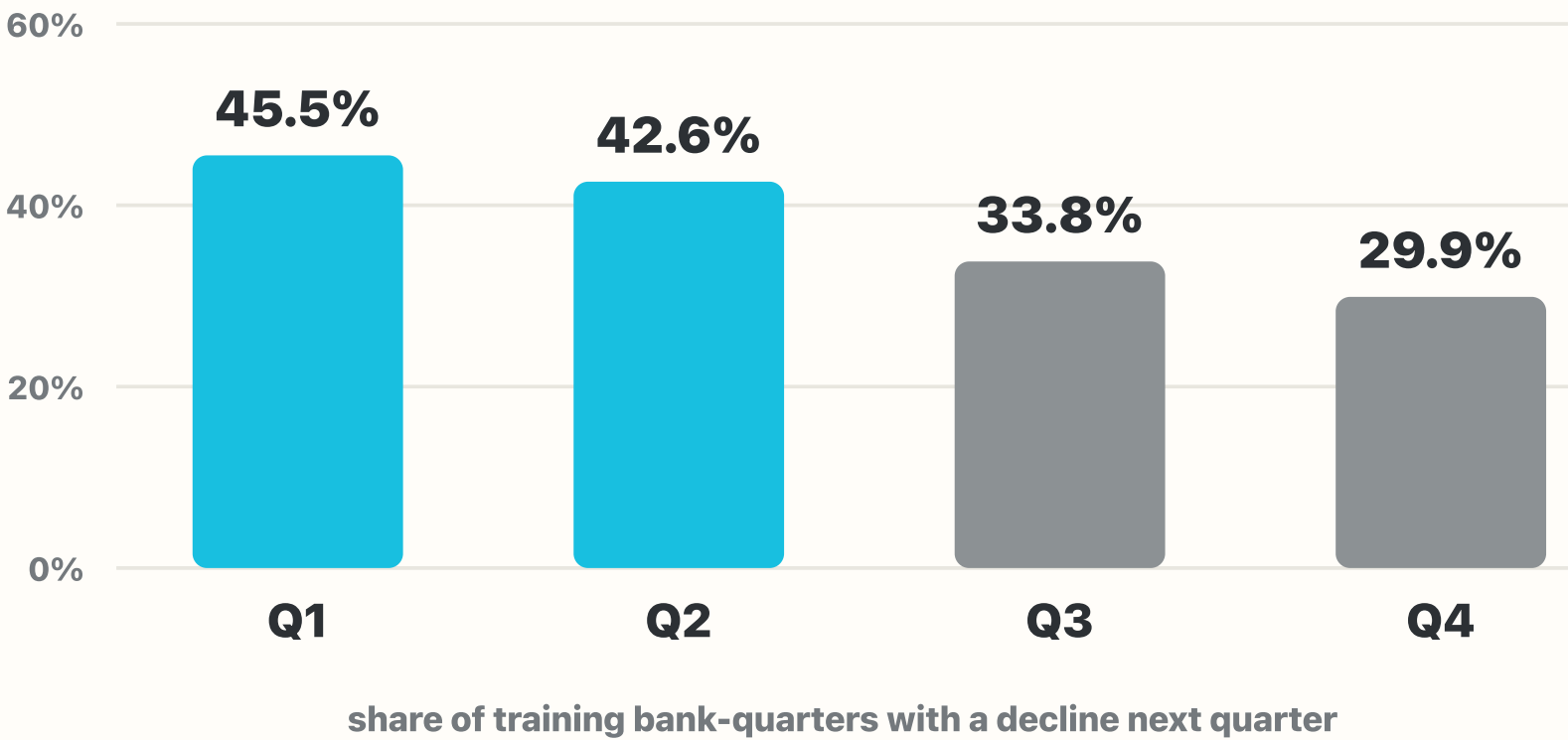
DEPOSIT SIZE ALONE: 18.8 FOUND PER 100 REVIEWED.
ALL FIVE INPUTS: 21.9. A GAP OF 3.1.

DEPOSIT DECLINES WERE **MORE COMMON AFTER Q1** THAN Q4.

Q1: 45.5% of training bank-quarters were followed by a deposit decline. **Q4: 29.9%.**

That is how often declines happened, not “banks lost 45.5% of their deposits.”

TRAINING DATA, 2013–2022 · BANK-QUARTERS POOLED BY STARTING QUARTER



THE STARTING QUARTER MAY HELP **SET A GROWTH FORECAST.**

Each bar pools all eligible bank-quarter rows for that starting quarter in the training set (through Sep 2022). Declines ÷ rows; not an equal-weight average of yearly rates.

MODEL CHOSEN FOR THE NEXT TEST: THE CALENDAR RULE.
NOW WE WAIT FOR Q3 REPORTS.

Q3 ends September 30, 2026. Reports arrive afterward. The saved forecast is +0.550% per bank; zero is the benchmark.

CALENDAR RULE · Q2 STARTING QUARTER

+0.550%

Same forecast for every bank. If actual growth is 0%, the absolute error is 0.550 pp.

NO-CHANGE BASELINE

0%

Predict no change. If actual growth is 0%, the error is zero.

1. WAIT FOR SEPTEMBER REPORTS.

Actual growth =
 $100 \times (\text{September deposits} \div \text{June deposits} - 1)$.

2. MEASURE EVERY ERROR.

Absolute error = $|\text{actual growth} - \text{forecast}|$.
Example: +1.00% actual → 0.450 pp calendar error; 1.000 pp zero error.

3. COMPARE THE AVERAGES.

Lower MAE wins this quarter.
If the calendar rule loses, report the loss. Inspect errors before changing the rule.

THE CALENDAR RULE WON HISTORICAL MAE: 3.500 VS 3.601.
WE'LL SEE WHETHER IT WINS THE NEXT QUARTER.

As of Sep 22, 2026: June data are listed in FDIC BankFind; the Sep 30 quarter has not ended. Saved rule = Q2 training median (+0.550399842%); developed after seeing 2024. Score matched banks with valid positive June and September domestic deposits; disclose missing reports and merger cases. Example is illustrative. One quarter is not lasting proof.